THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 05/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

Stakeholders struggle to visualize civil infrastructure impacts from 2D blueprints during public zoning reviews.

2. AI RATIONALE & ANALYSIS

"Mixed Reality enables non-technical stakeholders to intuitively comprehend 3D civil infrastructure scale and environmental context directly over spatial meeting environments. THEXRA leverages Microsoft HoloLens 2 to deliver synchronized collaborative reviews, significantly streamlining public zoning approvals."

3. PROPOSED SOLUTION CONCEPT

A multi-user Mixed Reality visualization solution that projects interactive 3D civil infrastructure models over physical topographic layouts and maps, allowing zoning review committees and the public to examine scale, environmental impact, and sightlines collaboratively in real time.

IMPLEMENTATION APPROACH:

1. Phase 1: Ingestion and optimization of municipal CAD/GIS blueprint data into real-time 3D models.
2. Phase 2: Deployment of multi-user spatial anchoring and interactive UI controls for zoning review meetings.
3. Phase 3: Acceptance testing, municipal stakeholder onboarding, and production zoning hearing rollout.

DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (1 week):Infrastructure CAD/GIS data audit report • Public zoning workflow technical requirements specification
2. **Design** (1.5 weeks):Spatial user interface wireframes • Multi-user zoning review interaction guide
3. **Prototype** (1.5 weeks):Single-model HoloLens 2 demonstration app • Initial spatial anchor alignment report
4. **Development** (3 weeks):Multi-user shared holographic zoning system • Layer-toggle and measurement tool module
5. **Testing** (1.5 weeks):Network latency and multi-user load test report • User experience validation results with planning staff
6. **Deployment** (1.5 weeks):Hardware provisioning and room configuration guide • Live zoning board pilot deployment
7. **Support** (Ongoing):Quarterly headset OS and firmware updates • Continuous technical support for scheduled public hearings

SCOPE OF WORK (SOW)

1. PROJECT SCOPE

Delivery of an interactive Multi-User Mixed Reality Zoning Review application converting civil infrastructure blueprints into synchronized 3D holographic models for up to 6 simultaneous participants.

2. CORE FEATURES

- Synchronized multi-user spatial hologram viewing
- Interactive civil layer controls (water, electrical, traffic, building envelopes)
- True-to-scale horizon and sightline impact inspection tool
- Voice and gesture spatial annotation logging

3. DELIVERABLES

- Configured Microsoft HoloLens 2 review kit
- Pre-loaded municipal zoning 3D spatial application
- Zoning facilitator control tablet app
- Administrator and operator manual

4. HARDWARE REQUIREMENTS

- 6x Microsoft HoloLens 2 Headsets
- 1x Dedicated Local Wi-Fi 6 Access Point
- 1x Facilitator Control Tablet

5. SOFTWARE MODULES

- THEXRA MR Spatial Zoning Application License
- Multi-User Synchronization Gateway License

6. CONTENT ASSETS

- Up to 3 converted and optimized 3D civil infrastructure project datasets
- Spatial contextual terrain model of the target zoning district

7. TRAINING PLAN

Two half-day hands-on facilitation training workshops for municipal urban planning staff and technical moderators.

8. DEPLOYMENT STRATEGY

On-site setup and calibration within the municipal civic council chamber,

9. ONGOING SUPPORT

3 months of post-deployment technical hearing assistance, system

including network optimization and spatial anchor baseline.

maintenance, and bug fixes.

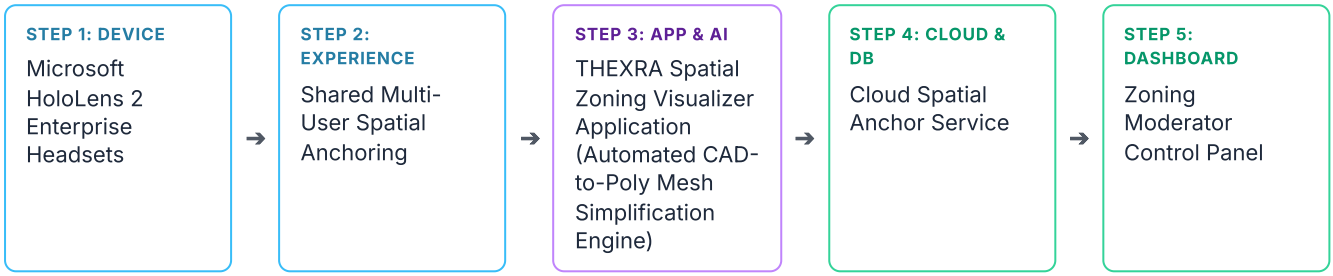
10. EXCLUSIONS (OUT OF SCOPE)

- Original architectural CAD/BIM model drafting or structural civil redesign
- Long-term civic hall electrical and audiovisual rewiring
- Mobile smartphone or consumer VR platform porting

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

MR — Microsoft HoloLens 2

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: The solution converts municipal CAD and GIS civil blueprints into real-time interactive 3D holographic digital assets. Stakeholders wear Microsoft HoloLens 2 headsets to engage with synchronized multi-user architectural overlays, while an interactive display mirrors the view for general audiences.

2. User Journey: Stakeholders join the zoning session and don the MR headset, automatically synchronizing to the shared spatial anchor over a physical meeting table. They inspect proposed infrastructure layers, toggle utilities and environmental data via hand gestures, and submit spatial annotations for the formal record.

3. Hardware: Microsoft HoloLens 2 Enterprise Headsets, Local Wi-Fi 6 Spatial Anchor Router, Presentation Display Monitor

4. Software: THEXRA Spatial Zoning Visualizer Application, Unity Mixed Reality Engine Runtime, Spatial Anchor Synchronization Service

5. AI Components: Automated CAD-to-Poly Mesh Simplification Engine, Natural Language Voice Command Processor

6. XR Components: Shared Multi-User Spatial Anchoring, Interactive 3D Layer Toggling (Utilities, Traffic, Foliage), True-to-Scale Sightline Occlusion Visualizer

7. Backend: Cloud Spatial Anchor Service, Secure Municipal GIS / CAD Asset Repository, Session Review Logging Database

8. Dashboard: Zoning Moderator Control Panel, Live Public Spectator Casting Screen, Stakeholder Feedback and Spatial Annotation Log

9. Data Flow: Civil GIS/CAD Data → Automated Polygon Optimizer → Cloud Spatial Service → HoloLens 2 Headsets → Live Spectator Display

Alternative Option: Interactive Display — THEXRA View

5. EXPECTED BUSINESS OUTCOMES

60% reduction in zoning review deliberation times, 80% improvement in non-technical stakeholder comprehension, and early identification of civic design disputes.

6. IMPLEMENTATION NEXT STEPS

Gather existing CAD/GIS blueprint datasets and run a spatial conversion technical validation workshop with the urban planning team.